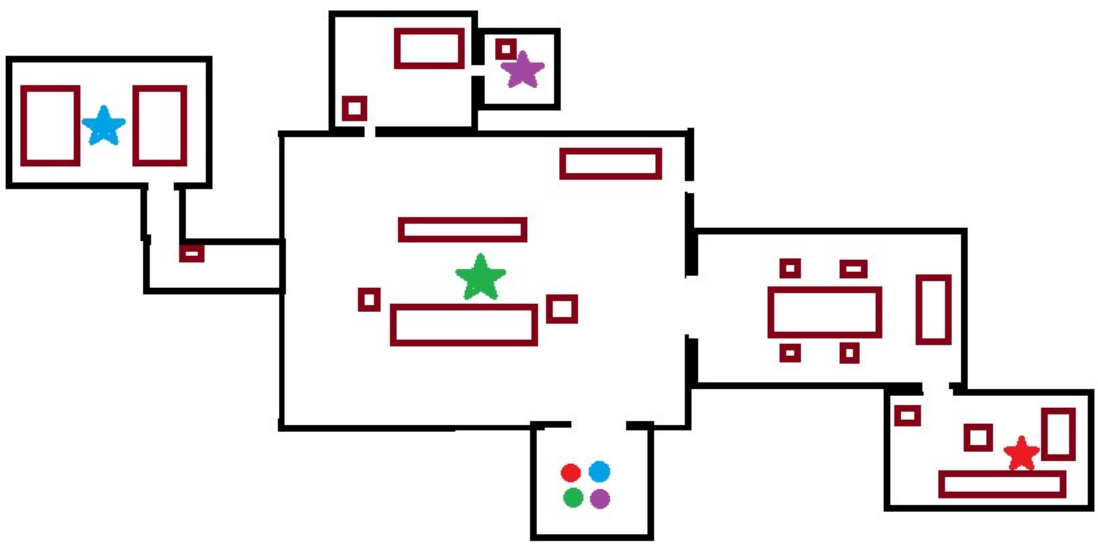
Game project Idea:

Scenario: Four NPCs are trying to get to different places in a house.

Goal: Test different kinds of path planning techniques

Scene:

- Different rooms (house)
- NPCs
  - Each NPC will come with an array of 2 – 3 targets they will visit
  - For each target they will do the following:
    1. Walk to target (State: WALKING)
    2. Interact with the target (State: IDLE)
       - Once the NPC reaches the target and enters the idle state, a timer will count down when they should start moving again
  - This process is continued for each target in the list, when the end of the array is reached, the process will loop.



Example of one target state

- Option list with at least three path planning algorithms (default will be A*)

- Obstacles
  - Maybe add a moving obstacles as well to add some complexity

Basic animation key frames:

Algorithms to implement

- A*
- Maybe D* (professor said it was too complex to detail in lecture so might be sophisticated enough)
- Particle Repulsion

Techniques from class:

- Portal Collision Detection
- Key frame animation
- Finite State machine (idle, walking, interacting)